DISSERTATION INTRODUCTION

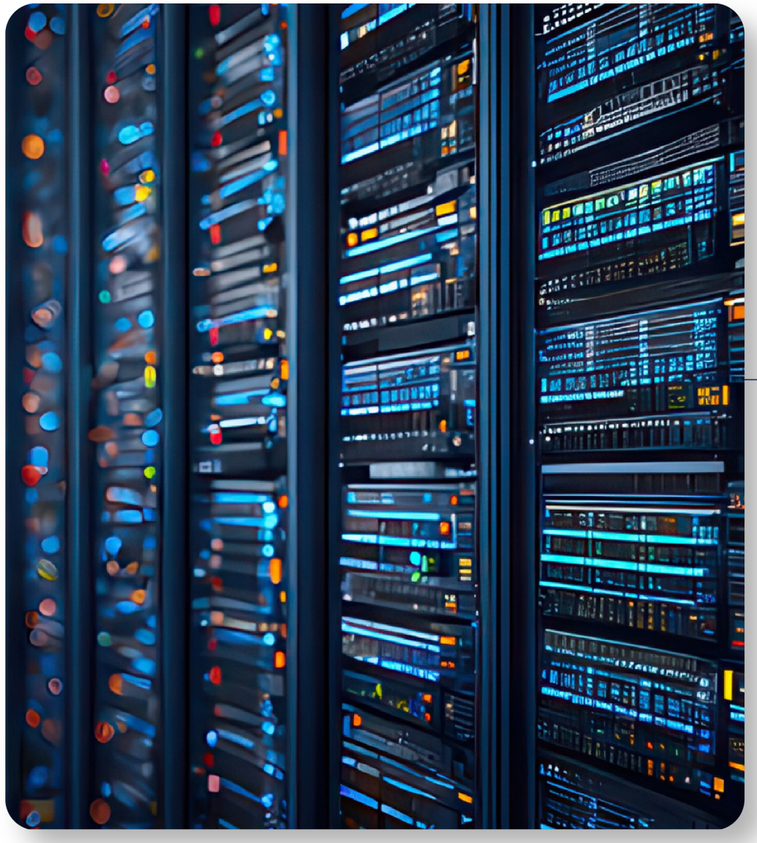
Hallucination Across various Retrieval Augmented Generation (RAG) Strategies

A comparative study on how different RAG architecture influence LLM hallucinations.

ANKIT MAHTO - 13454

INTRODUCTION

- In modern times, advancement in Artificial intelligence has given birth to large language models have been revolutionary across all aspects of human life.
- Large language models are being adopted in various fields such as medical, banking, retail, corporate, academia...
- However, these LLMs have some underlying issues, most notable being LLM hallucinations.
- This is where RAG comes in, in recent years, RAG has proven to be a great method for mitigating LLM hallucinations along with other problems.
- This research paper will compare and contrast different architectures of Retrieval-Augmentation Generation to find correseponding hallucination rates in large language models (LLMs).
- This research paper will be descriptive/exploratory in nature.



BACKGROUND

1

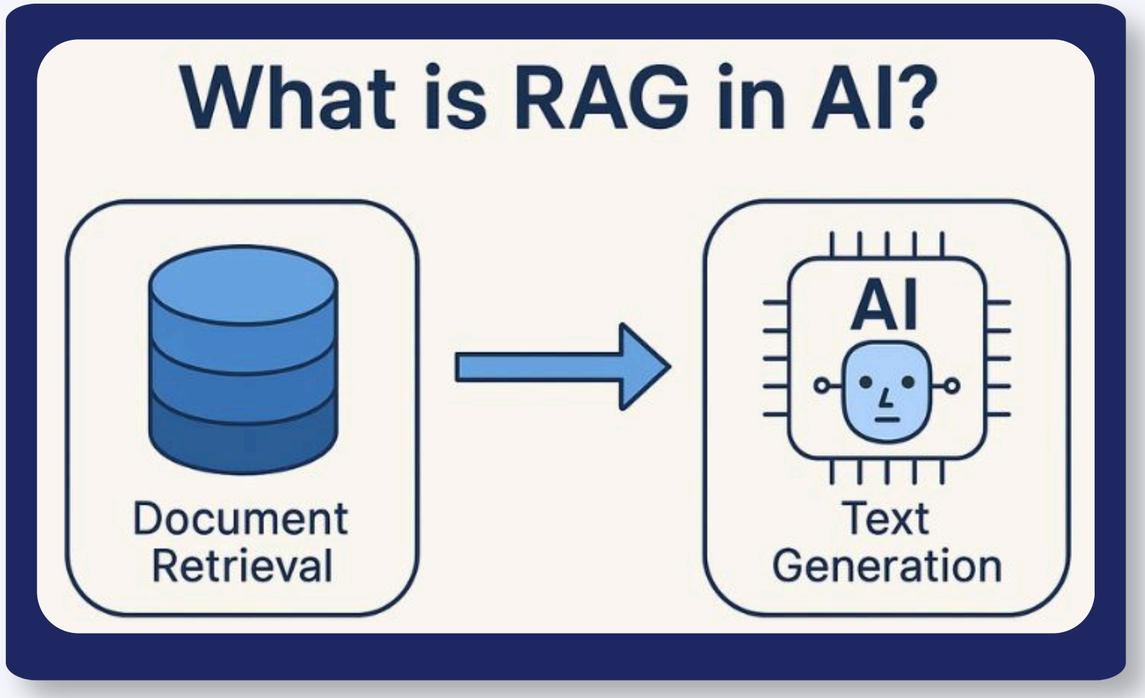
What is retrieval-augmentation generation (RAG)

RAG is a AI architecture by which a LLM model can access private data.

2

Why do we need RAG?

- A LLM is pretrained models that predicts the ouput based on the parametric knowledge (pretrained knowledge).
- A LLM by default doesn’t have access to outside knowledge
- This leads to several issues, which we will see further



Problems RAG solves with plain LLM

Common issues encoutered when dealing with vanilla LLM

1

Knowledge cut-off

LLMs don’t have the knowledge of newere data which they werent trained on. so they can’t answer questions related to recent events.

2

Lacking private data

LLMs dont have access of the personal data. Therefore, they can’t answer questions related to personal data

3

Expensive training

Training LLMs with newer data takes time and cost. RAG solves this by feeding newer knowledge to LLMs directly

4

Hallucinations

When LLMs don’t have the required data, they hallucinate and make up rnadom information and reply confidently. This is called hallucination effect

These are some of the core problems that RAG solves effectively. But for the sake of this study we will focus on hallucination part.

Purpose of this study

Why we are doing this study

01 **Find hallucination rates accross different RAGs**

02 **Compare and contrast accross different sub fields**

03 **Literature gap**

Literature review

S.no.	Paper	Publisher	Subject	Outcome
1	Maximizing RAG efficiency: A comparative analysis of RAG methods	Cambridge University Press: 30 October 2024		
2	EvaRAG: Evaluating Advanced RAG Techniques With Indexing and Distance Metrics	IEEE		
3	TYPED-RAG: Type-Aware Decomposition of Non-Factoid Questions for Retrieval-Augmented Generation	Association for Computational Linguistics		
4	Re2G: Retrieve, Rerank, Generate	Association for Computational Linguistics		
5	A Survey on RAG with LLMs	Science Direct		
6	A survey on retrieval-augmentation generation (RAG) models for healthcare applications	Springer	Compares types of rag in Medical Field	
7	A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications	IEEE	Compares different retreival types	
8	Optimising clinical information extraction: a comparative study of retrieval-augmented generation techniques in clinical notes	Science Direct	Compares different retreival types in medical field	
9	Evaluating Sparse and Dense Retrieval in Retrieval-Augmented Generation Systems: A Study	Association for computing machinery (ACM)	Compares different retreival types	
10				

RAG Taxonomy

Retrieval decides what the model even gets to see

1

Query

A question comes in, needing information beyond the model's parameters.

2

Retrieve

An index (vectors, keywords, or a graph) pulls candidate passages from a corpus.

3

Augment

Retrieved passages are inserted into the model's context window.

4

Generate

The LLM writes an answer — ideally grounded in what it was handed.

Basic architecture of RAG

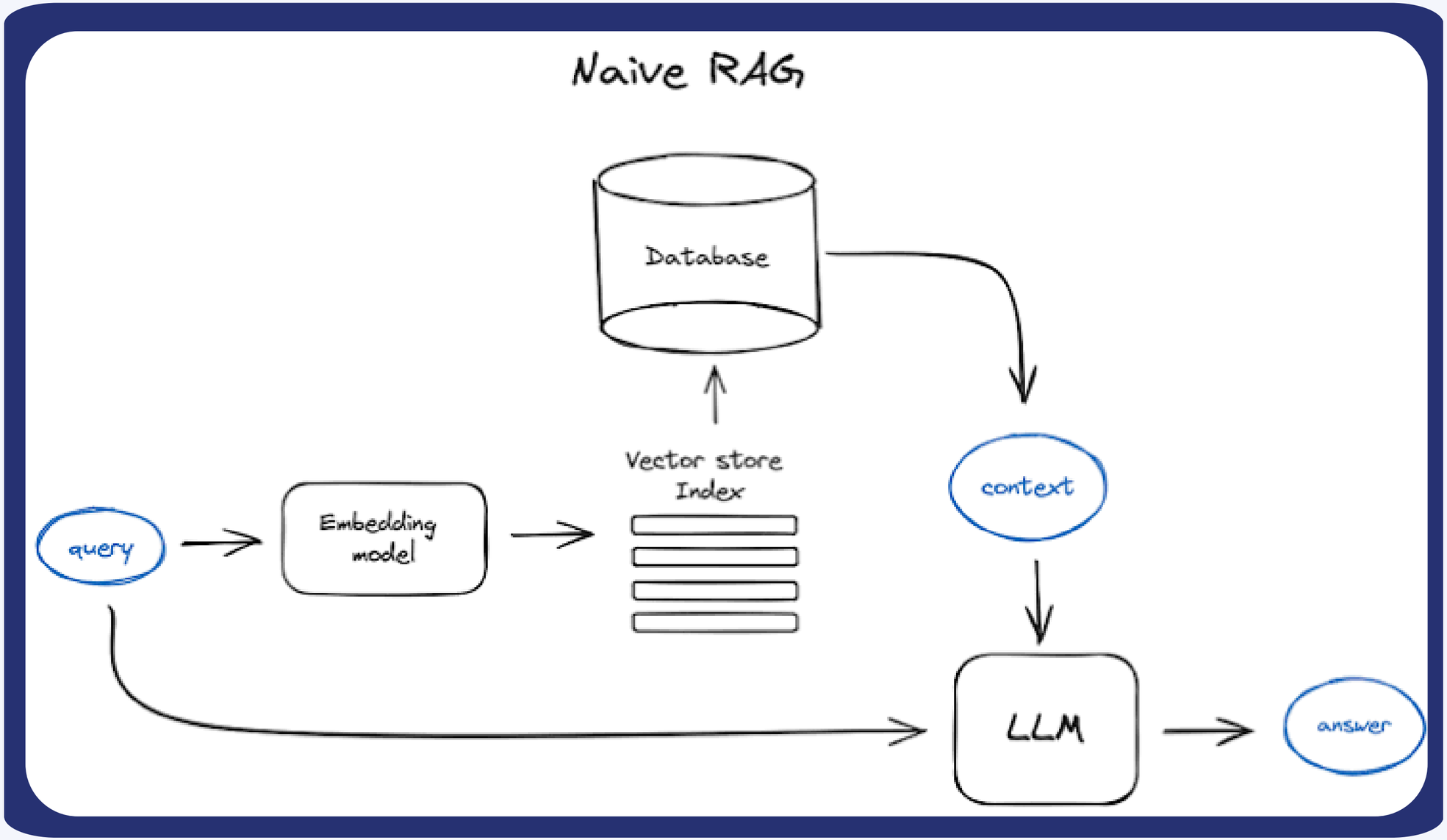
RAG Components

Indexing

Retrieval

NOTE: for the sake of this study, we will only only consider two major components of RAG as these are where most changes and comparisons would be made. Other parts such as Augmentation and generation will mostly remain constant

Naive RAG Architecture



Other RAG Architectures

Naive RAG

Advanced RAG

Graph RAG

Modular RAG

- Self-RAG
- Corrective RAG
- Agentic RAG
- Adaptive RAG

Methodology

Proposed mehtodology for this study



Datasets

Datasets that can be used for retrieval

Medical

- MedRAG
- Mirage
- pubMedQA

Law

- Legalbench-RAG
- SC Judgements
- Legal RAG bench

Corporate

- EnterpriceRAG bench
- RAG multi corpus
- Hugging face datasets

Science

- SciQ
- RAGCare-QA
- Open RAG bench

LLM Models

Top Open source models

KIMI K3

Parameters: 2.8 Trillion
Context length: 1 million token
Disk size: 1.56 TB

GLM 5.2

Parameters: 753 Billion
Context length: 1 million token
Disk size: 1.51 TB

Gemma 4 26B A4B

Parameters: 26 Billion
Context length: 256K token
Disk size: 48 GB

Deepseek V4 Pro

Parameters: 1.6 Trillion
Context length: 1 million token
Disk size: 893 GB

Qwen 3.8-Max

Parameters: 2.4 Trillion
Context length: 1 million token
Disk size: 1.2 TB

Llama 4 Maverick

Parameters: 400 Billion
Context length: 1 million token
Disk size: 800 GB

Min-Max M3

Parameters: 428 Billion
Context length: 1 million token
Disk size: 128 GB

Things to consider

Changes can be applied to different components

Core Architecture

- Naive RAG
- Advanced RAG
- Modular RAG
- Agentic RAG

Retrieval mechanism

- Sparse (BM25)
- Dense (embeddings)
- Hybrid (fused)

Knowledge structure

- Flat / chunked
- Hierarchical (RAPTOR)
- Graph RAG

Control flow/behavior

- Self-RAG
- Corrective RAG
- FLARE
- Adaptive RAG

Things to consider

Changes can be applied to different components

RAG EVALS

Score the whole pipeline

- Context precision — are retrieved chunks relevant?
- Context recall — was everything needed retrieved?
- Faithfulness — does the answer match retrieved context?
- Answer relevance — does it address the query?

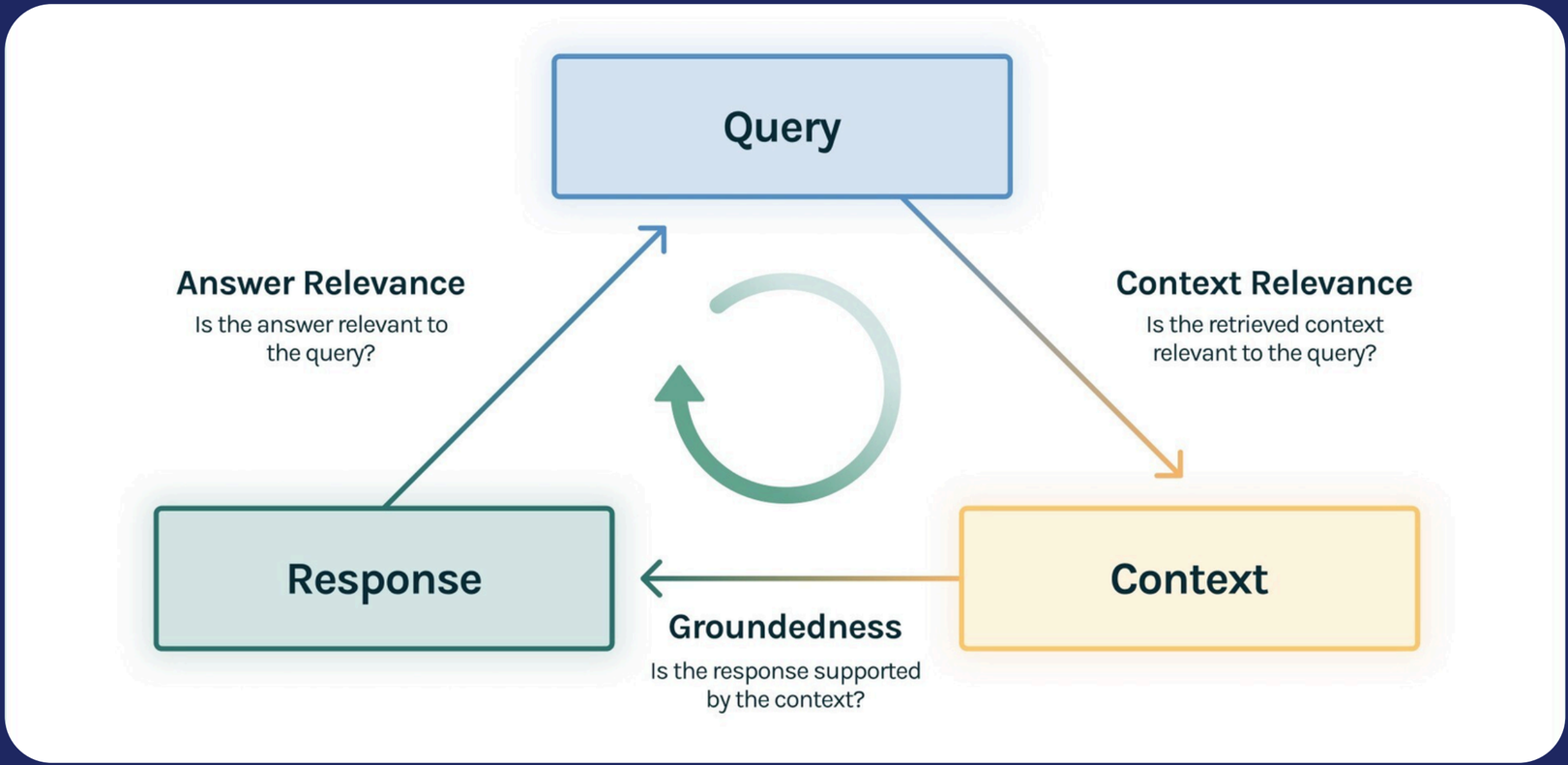
HALLUCINATION EVALS

Score the output on its own terms

- TruthfulQA — common-misconception factuality
- HaluEval — paired correct / hallucinated samples
- SelfCheckGPT — consistency across samples
- RAGTruth — word-level hallucination spans in RAG

Evals

Evaluation Metric: RAG Triad



NOTE: Hallucination on in itself cannot be measured by just single metric, thats why we will be looking at multiple factors to judge the outcome. The best way to judge this is by using RAG traid. It measures the out put based on 3 metrics: Context Relevance, Answer Relevance and groundedness. It does 2 test before output generation and 2 after the generation.

Eval Metrics

Retrieval

1

Context recall

Determines if all necessary pieces of information were successfully retrieved by the system

2

Context relavance

Assesses whether the retrieved chunks contain relevant information with minimal noise.

Generation

3

answer relevance

Measures how directly the final answer addresses the user's initial query.

4

Faithfulness

Checks if the generated response is strictly grounded in the retrieved context to reduce hallucinations.

Evals

Potential Eval frameworks

RAGAS

DeepEval

LangSmith

September

Literature review

S.no.	Paper	Publisher	Subject
1	A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications	IEEE	Compares different RAGs on different fields
2	EvaRAG: Evaluating Advanced RAG Techniques With Indexing and Distance Metrics	IEEE	Compares different RAG eval frameworks
3	Maximizing RAG efficiency: A comparative analysis of RAG methods	Cambridge University Press: 30 October 2024	Compares RAGs against Accuracy vs Efficiency
4	Re2G: Retrieve, Rerank, Generate	Association for Computational Linguistics	Optimizing and comparing RAGs
5	A Survey on RAG with LLMs	Science Direct	Survey compares different RAGs against Accuracy vs Efficiency
6	Evaluating Sparse and Dense Retrieval in Retrieval-Augmented Generation Systems: A Study	Association for computing machinery (ACM)	Compares different retreival types
7	Enhancing the Precision and Interpretability of Retrieval-Augmented Generation (RAG) in Legal Technology: A Survey	IEEE	Compares RAGs against different metrics
8	Evaluating Retrieval-Augmented Generation Variants for Clinical Decision Support: Hallucination Mitigation and Secure On-Premises Deployment	MDPI	Compares Rags on Medical fields and optimizes to reduce hallucination during deployment
9	A survey on retrieval-augmentation generation (RAG) models for healthcare applications	Springer	Compares Rags on Medical Datasets
10	Reducing Hallucinations and Trade-Offs in Responses in Generative AI Chatbots for Cancer Information: Development and Evaluation Study	International Journal on Engineering Artificial Intelligence Management, Decision Support,	Compares different RAGs on medical field
11	Optimising clinical information extraction: a comparative study of retrieval-augmented generation techniques in clinical notes	Science Direct	Compares different retreival types in medical field
12	LegalGraphRAG: Multi-Agent Graph Retrieval-Augmented Generation for Reliable Legal Reasoning	ACL	LEGALGRAPH RAG for legal field
13	Towards Reliable Retrieval in RAG Systems for Large Legal Datasets	ACL	Summary-Augmented Chunking (SAC) for Legal documents
14	Bridging Legal Knowledge and AI: Retrieval-Augmented Generation with Vector Stores, Knowledge Graphs, and Hierarchical Non-negative Matrix Factorization	ACM	RAG adaption for Legal Field
15	Hallucination in Retrieval-Augmented Generation is a Context Construction Problem: A Constraint-Driven Cross-Modal Framework	ACM	Constraint driven RAG framework for construction field
16	MEGA-RAG: a retrieval-augmented generation framework with multi-evidence guided answer refinement for mitigating hallucinations of LLMs in public health	Frontiers	MEGA Rag for public health domain
17	Claim-Aware Scientific RAG: Evidence-First Retrieval and Abstention for Scientific Fact Responses on SciFact	Journal of Advanced Computing Systems (JACS)	Claim aware retrieval
18	Retrieval Augmentation Reduces Factual Errors in Knowledge-Intensive Language Model Tasks	Computer Life	Simple RAG with Eval
19	TYPED-RAG: Type-Aware Decomposition of Non-Factoid Questions for Retrieval-Augmented Generation	Association for Computational Linguistics	Provides new RAG for making system question type aware

Literature Gap

1

Fewer RAG architecture comparisons

Scouting the global research databases, i’ve found that there are very few studies that compare RAG architechture for hallucination and other metrics

2	EvaRAG: Evaluating Advanced RAG Techniques With Indexing and Distance Metrics	IEEE	Compares different RAG eval frameworks
3	Maximizing RAG efficiency: A comparative analysis of RAG methods	Cambridge University Press: 30 October 2024	Compares RAGs against Accuracy vs Efficiency
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Literature Gap

SciFact

TriviaQA

LegalBench

SQuAD

PubmedQA

MedRag

2

Studies keep datasets constant

Studies tend to keep the datasets constant, often using datasets from a same field throuout the study

Literature Gap

3

Many comparison studies in medical field

Found many RAG architecture studies in the medical field using medical datasets such as MedRag & PubmedQA.

8	Evaluating Retrieval-Augmented Generation Variants for Clinical Decision Support: Hallucination Mitigation and Secure On-Premises Deployment	MDPI	Compares Rags on Medical fields and optimizes to reduce hallucination during deployment
9	A survey on retrieval-augmentation generation (RAG) models for healthcare applications	Springer	Compares Rags on Medical Datasets
10	Reducing Hallucinations and Trade-Offs in Responses in Generative AI Chatbots for Cancer Information: Development and Evaluation Study	International Journal on Engineering Artificial Intelligence Management, Decision Support, and Policies	Compares different RAGs on medical field
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Literature Gap

12	LegalGraphRAG: Multi-Agent Graph Retrieval-Augmented Generation for Reliable Legal Reasoning	ACL	LEGALGRAPH RAG for legal field
13	Towards Reliable Retrieval in RAG Systems for Large Legal Datasets	ACL	Summary-Augmented Chunking (SAC) for Legal documents
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4

No comparisons in Legal RAGs

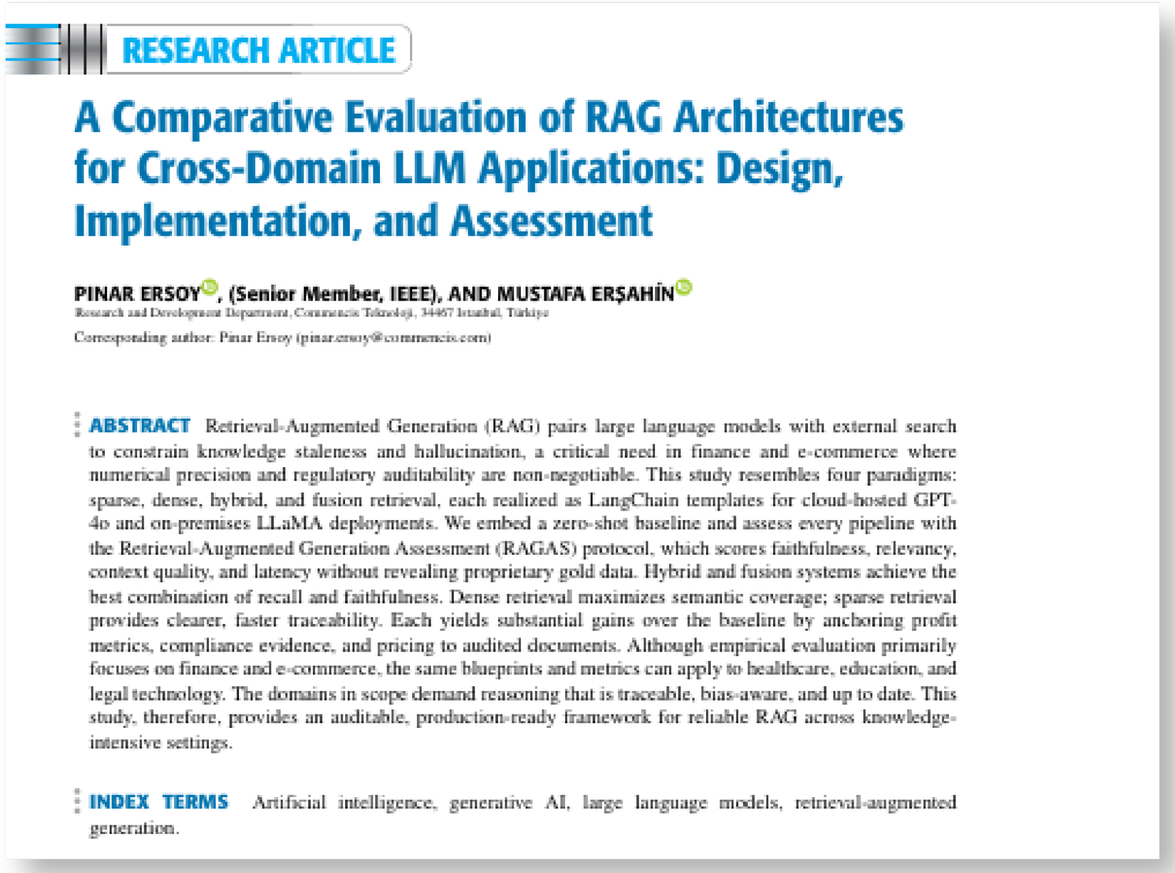
Legal RAGs tend to made around for optimization. So its hard to find studies that compare different architectures again legal datasets but rather try to optimize for legal field

Literature Gap

5

Only found 1 study that goes cross fields.

Only found a single study that goes cross fields in terms of using datasets while making comparisons and using different RAG architectures.



RESEARCH ARTICLE

A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications: Design, Implementation, and Assessment

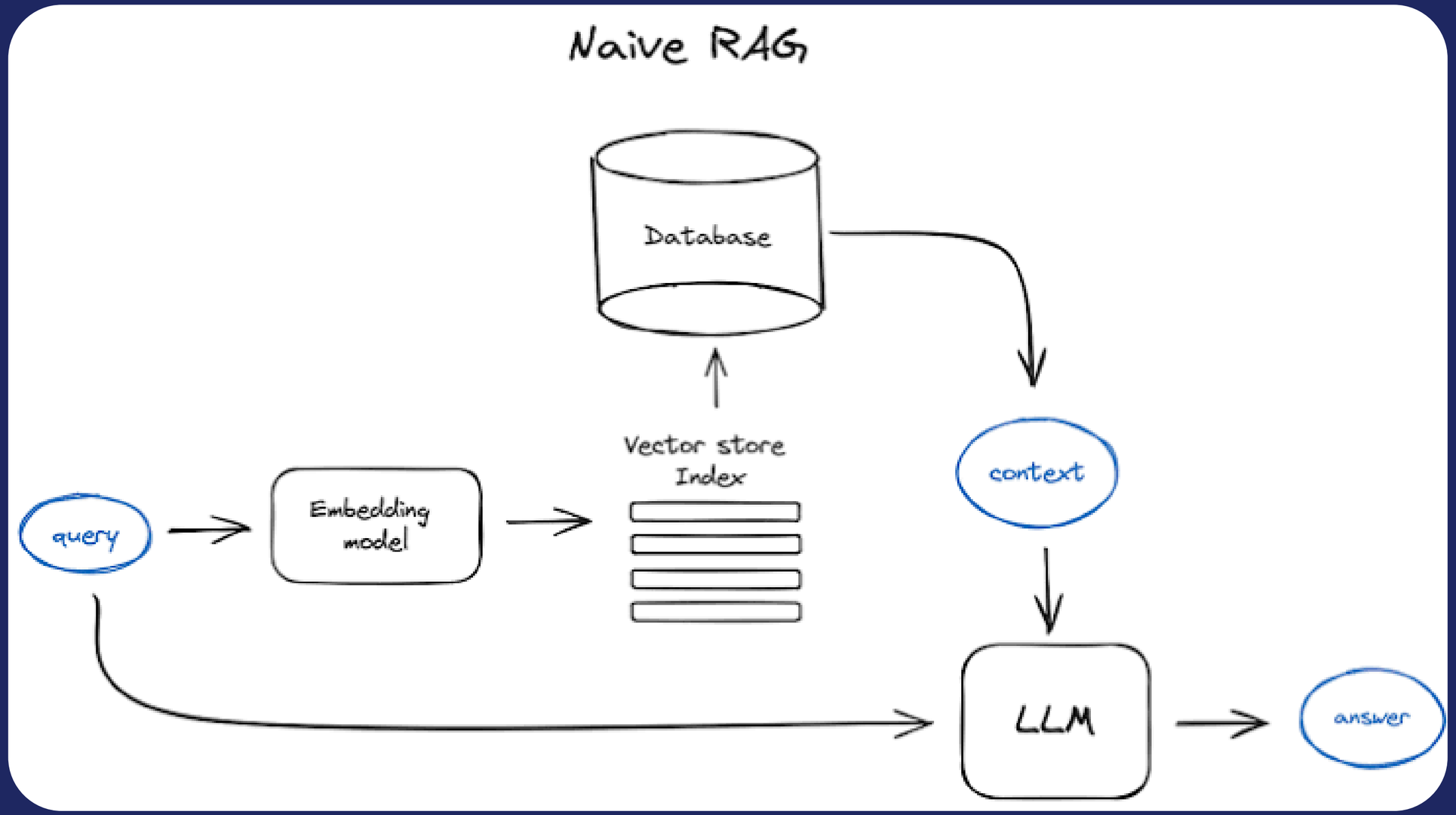
PINAR ERSOY, (Senior Member, IEEE), AND MUSTAFA ERŞAHİN
Research and Development Department, Commencis Teknoloji, 34467 Istanbul, Türkiye
Corresponding author: Pinar Ersoy (pinar.ersoy@commencis.com)

ABSTRACT Retrieval-Augmented Generation (RAG) pairs large language models with external search to constrain knowledge staleness and hallucination, a critical need in finance and e-commerce where numerical precision and regulatory auditability are non-negotiable. This study resembles four paradigms: sparse, dense, hybrid, and fusion retrieval, each realized as LangChain templates for cloud-hosted GPT-4o and on-premises LLaMA deployments. We embed a zero-shot baseline and assess every pipeline with the Retrieval-Augmented Generation Assessment (RAGAS) protocol, which scores faithfulness, relevancy, context quality, and latency without revealing proprietary gold data. Hybrid and fusion systems achieve the best combination of recall and faithfulness. Dense retrieval maximizes semantic coverage; sparse retrieval provides clearer, faster traceability. Each yields substantial gains over the baseline by anchoring profit metrics, compliance evidence, and pricing to audited documents. Although empirical evaluation primarily focuses on finance and e-commerce, the same blueprints and metrics can apply to healthcare, education, and legal technology. The domains in scope demand reasoning that is traceable, bias-aware, and up to date. This study, therefore, provides an auditable, production-ready framework for reliable RAG across knowledge-intensive settings.

INDEX TERMS Artificial intelligence, generative AI, large language models, retrieval-augmented generation.

Naive Rag

Base RAG Architecture



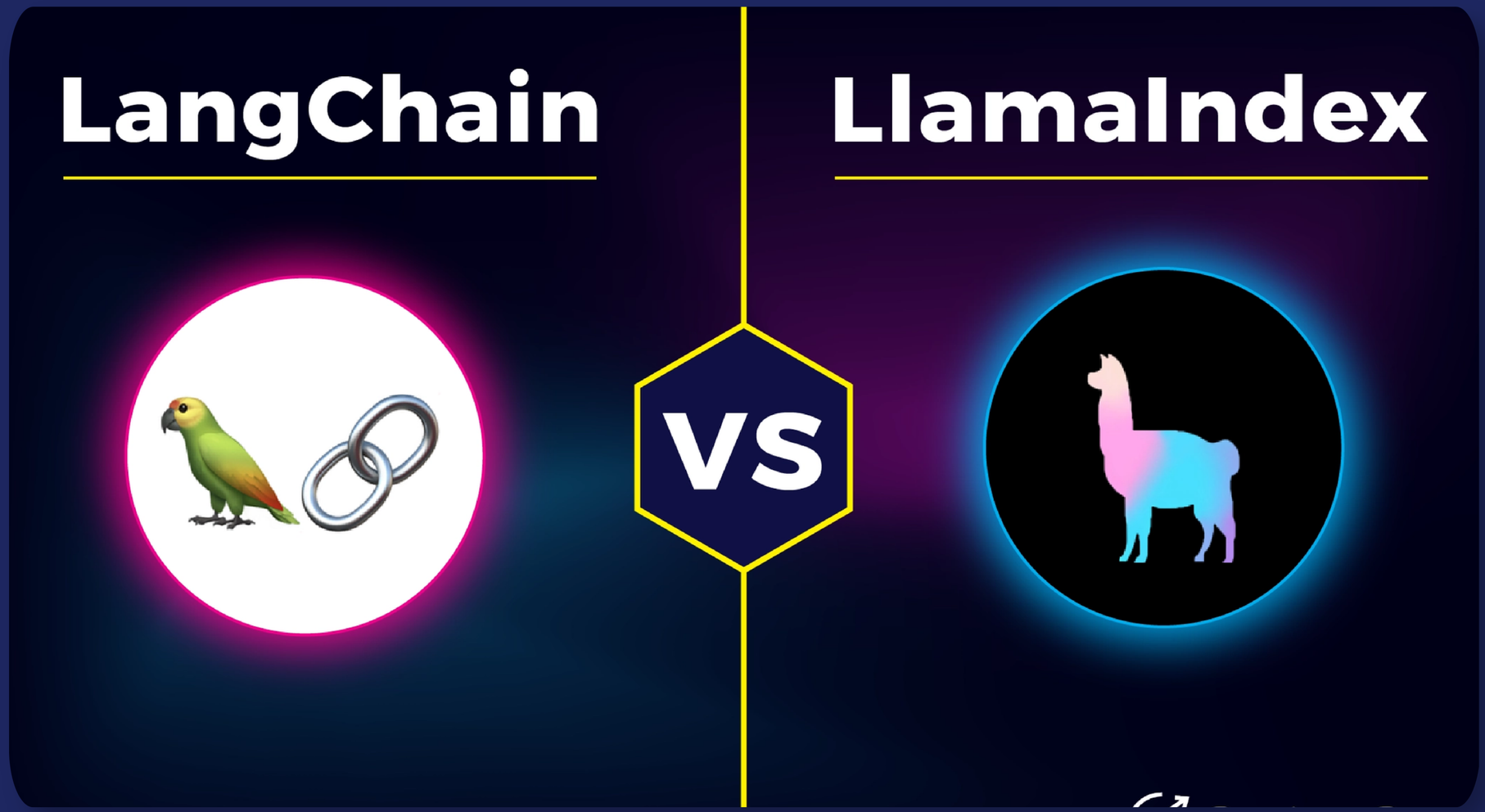
Framework

Base RAG Architecture



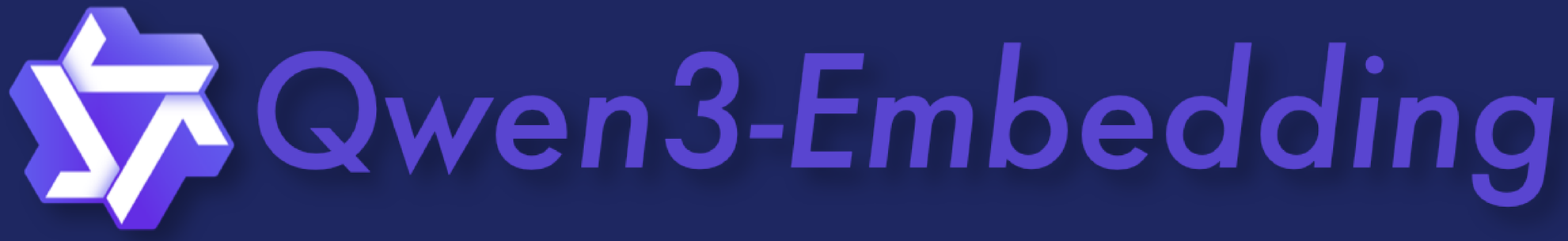
Why not LangChain?

Base RAG Architecture



Embeddings models

Base RAG Architecture





Vector Database

Base RAG Architecture

Default LlamaIndex Vector DB
simplevectorstore()



For this project, the defaul LlamaIndex vector store is enough given the datasets, but pinecornDB can also be used if the datasets size gets very very large.

LLM Models

Base RAG Architecture





Datasets used this month

Base RAG Architecture

Literature review

Due to hardware constraints, i’ve used the literature review on this subject as a dataset itself

SciQ dataset

A collection of 13,679 crowdsourced science exam questions covering physics, chemistry, biology, and other scientific fields

Data Preprocessing

Base RAG Architecture

Since i’ve used datasets in PDF format, we will need to first convert this into UTF-8 text format so the embeddings model can store it semantically and push to LLM for final query

1

Choose the PDF file

2

Run OCR to convert scanned pdf into editable text

3

Convert UTF-16 format to UTF-8

4

Push to RAG architecture

October

Recap

Previous Month

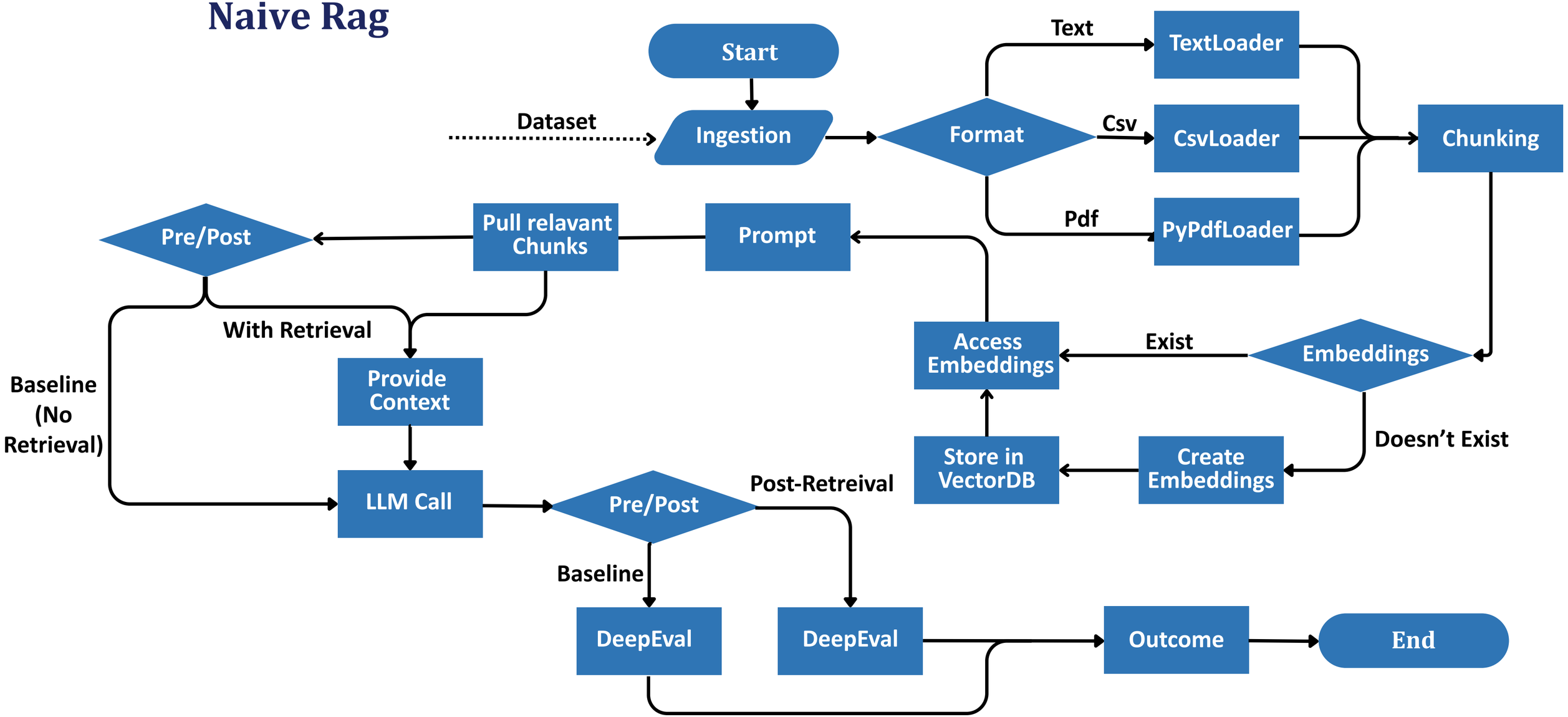
- We scoped out literature review.
- Potential concerns regarding datasets being used.
- Partial working prototype.

This Month

- Created working prototype of Naive RAG using Langchain.
- Added Environmental Dataset.
- Tested on all the datasets from various fields.
- Derived outcomes and noted results.

Workflow

Naive Rag



Outcome

Architecture	Datasets		Evaluation					
	Field	Dataset	Pre Retrieval - Baseline			Post Retrieval		
			HM	AR	FF	HM	AR	FF
Naïve RAG	Medical	WHO Covid	0	1	0	1	1	1
	Legal	Indian Legal Corpora Lora	0	1	1	1	1	0.5
	Corporate	Enterprise RAG Bench	0	0.75	0	1	1	1
	Environment	IPCC Dataset	0	0.846	1	1	1	1
	Science	UNEP Dataset	1	1	1	0	1	1
	General	20 News Group Dataset	0	0.8	0	0	0.5	0.25

How does DeepEval work?

Deep Eval Architecture

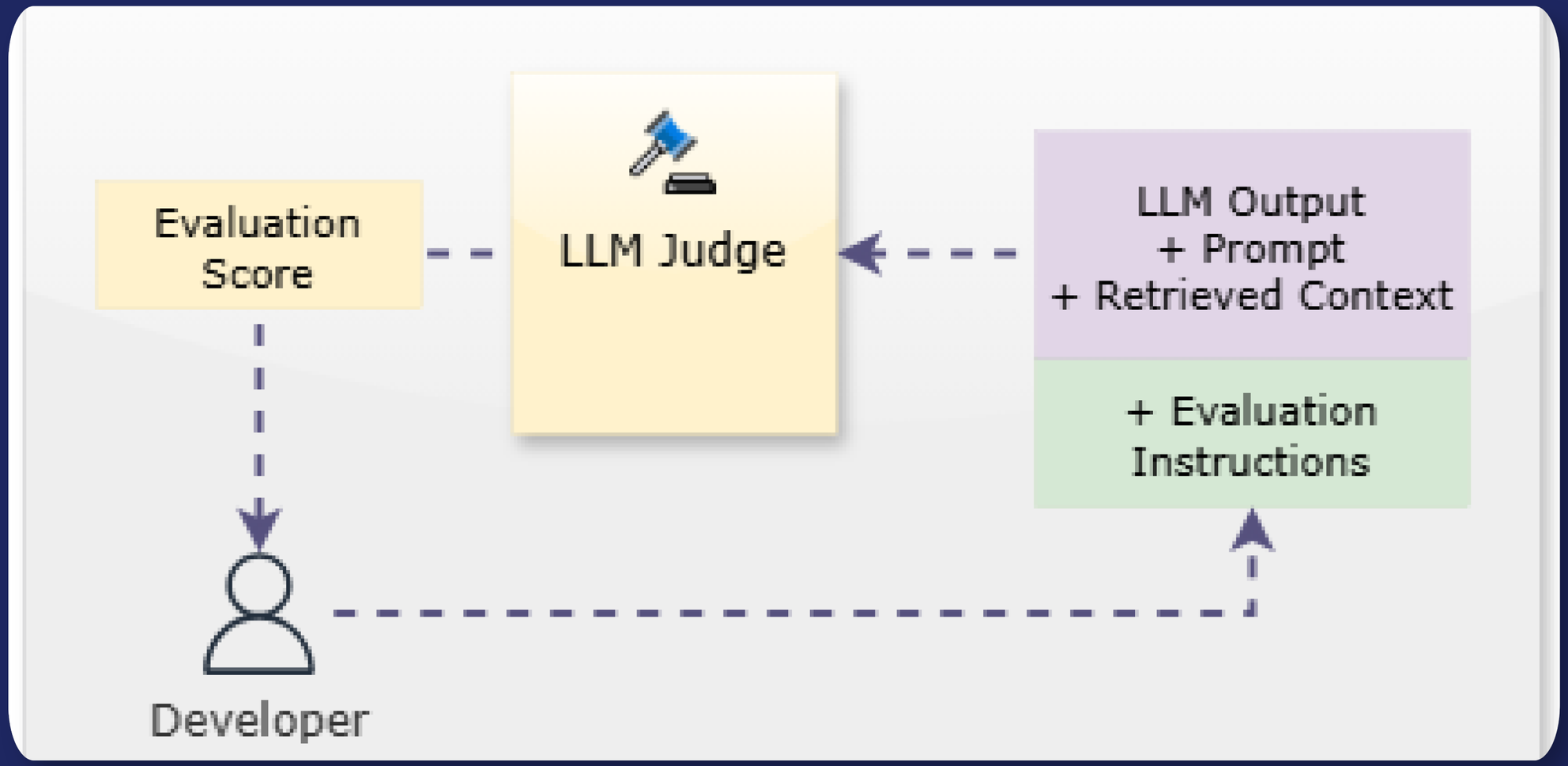
LLM as Judge

a technique that uses a large language model (LLM) to evaluate, score, or compare the text output of another AI system or agent against set standards



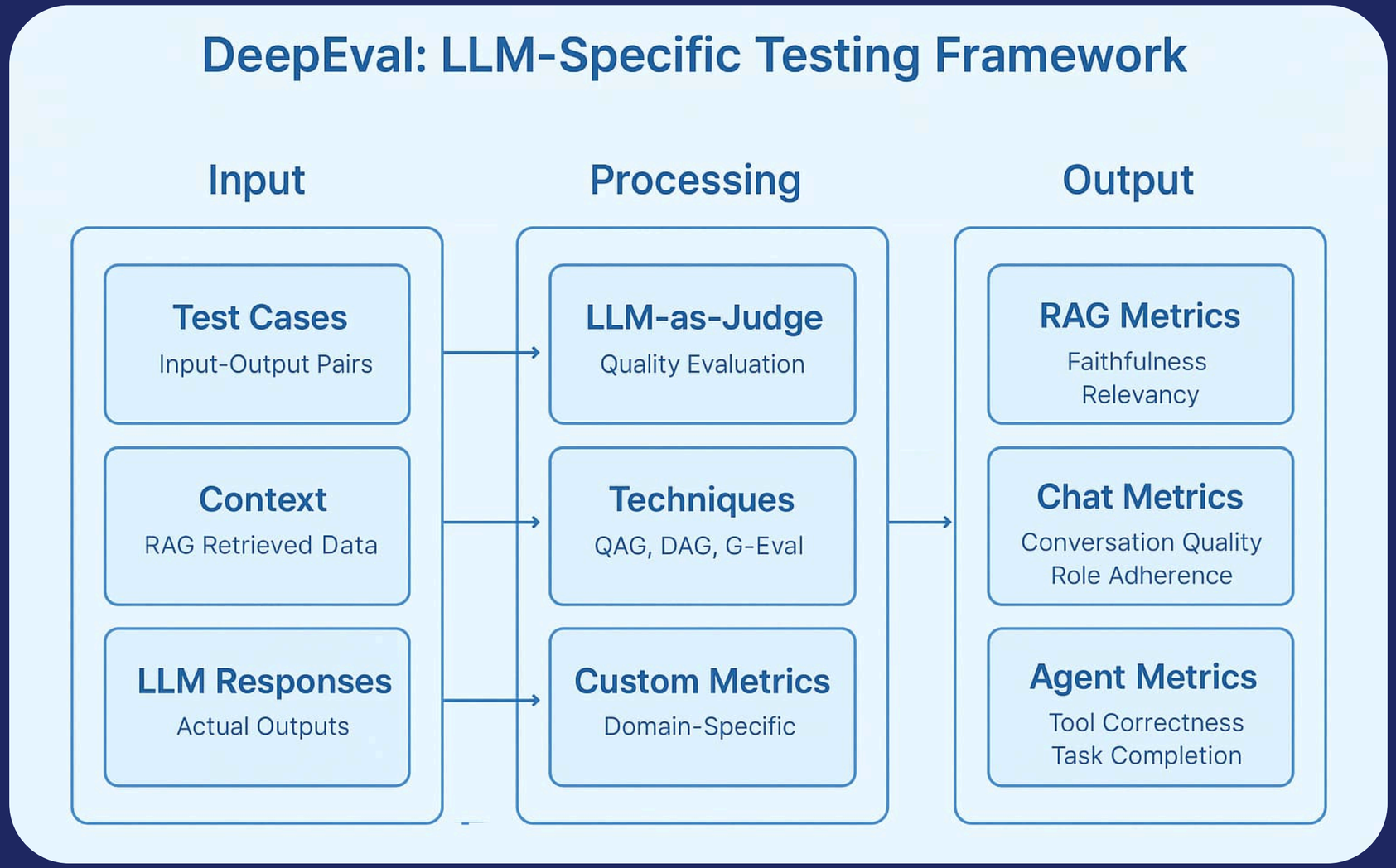
LLM as a Judge

Deep Eval Architecture



How does DeepEval work?

Deep Eval Architecture



Note

Deep Eval Architecture

**Since Judge is also a LLM, there's always a possibilty that verdict itself might hallucinate.... So to be at a safe place its generally recommended to use Judge LLM with roughly 2-3x more parameteres then generator LLM.
The following models below were choosen due to hardware constraints and that they fit well on the available hardware.**

Generator LLM

Qwen 3.5: 2b
2 Billion Parameters

Judge LLm

Gemma 4: e4b
4 Billion Parameters

Metrics

Deep Eval Architecture

1

Hallucination Metric

The HallucinationMetric in DeepEval is a reference-based metric that uses an LLM-as-a-judge to check whether a language model generates factually correct information by comparing its actual_output against a provided context.

2

Answer Relavency

An evaluation measure in generative AI and Retrieval-Augmented Generation (RAG) pipelines that scores how directly and appropriately a model's output addresses a user's input question.

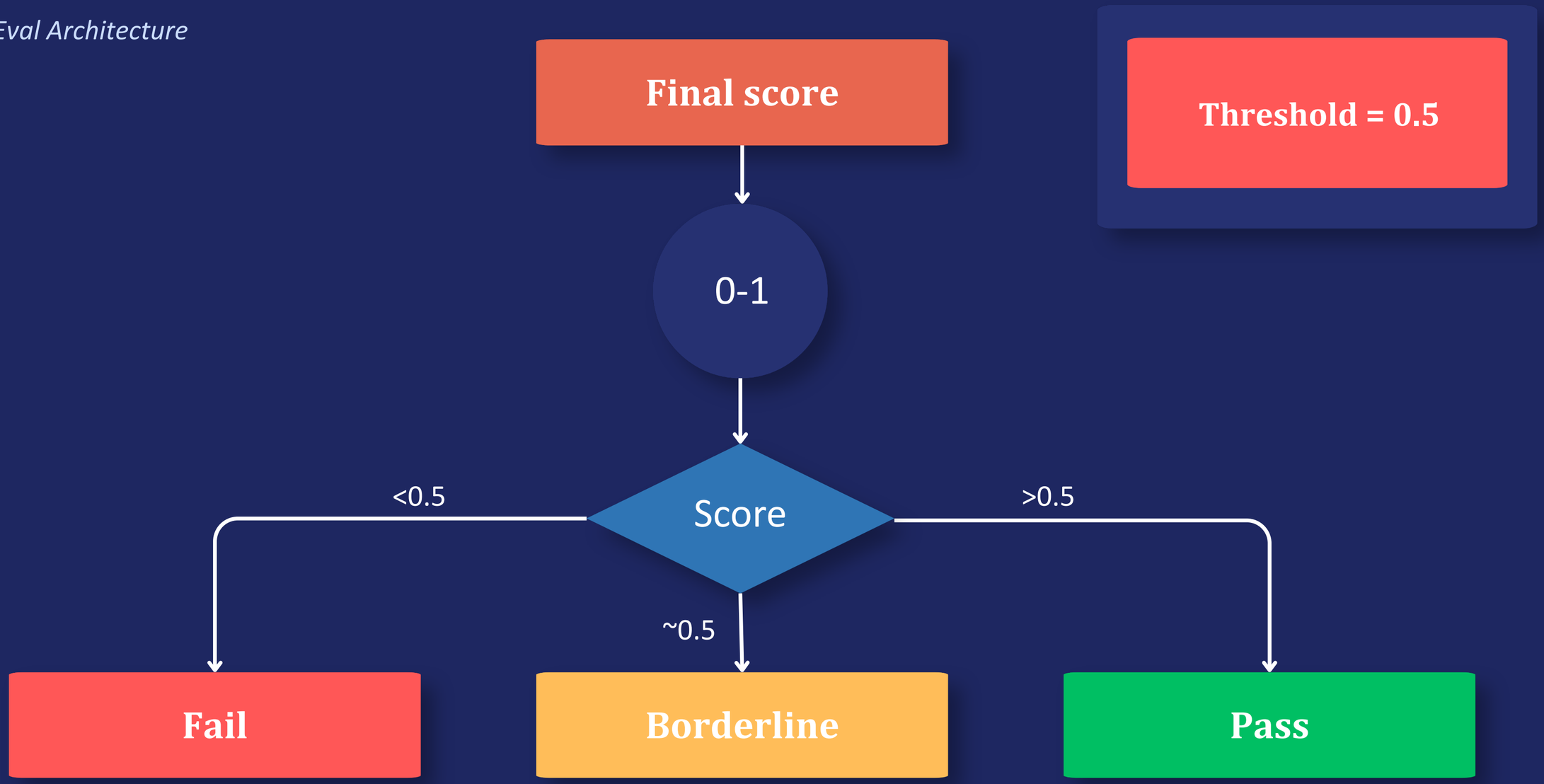
3

Failtfullness Metric

The Faithfulness Metric measures how factually consistent and grounded a generated AI response is relative to its retrieved context in Retrieval-Augmented Generation (RAG) systems

Score

Deep Eval Architecture



Outcome

Architecture	Datasets		Evaluation					
	Field	Dataset	Pre Retrieval - Baseline			Post Retrieval		
			HM	AR	FF	HM	AR	FF
Naïve RAG	Medical	WHO Covid	0	1	0	1	1	1
	Legal	Indian Legal Corpora Lora	0	1	1	1	1	0.5
	Corporate	Enterprise RAG Bench	0	0.75	0	1	1	1
	Environment	IPCC Dataset	0	0.846	1	1	1	1
	Science	UNEP Dataset	1	1	1	0	1	1
	General	20 News Group Dataset	0	0.8	0	0	0.5	0.25

Observations

1

Clear Improvement

- A clear improvemnt in hallucination metric can be observed when comparing to basline vs post retrieval outcome.
- The model generated better results when asked a factual question from the given dataset post retrieval.

2

Retrieval Quality

- It is observed that when the model is supplied with inadequate context, its much more likely to hallucinate.
- For instance, the eval on science dataset (UNEP) performed badly post retrieval but passed the test on baseline. This is because the model was trained on this dataset already and when supplied inaccurate context, it failed the test

3

Hallucination vs Faithfulness

- Theres a direct correlation between hallucination metric and faithfulness metric.
- If model performs/fails on faithfulness metric, its much more likely to hallucinate on the actual output.

Thank you